

***Oprl1* is the most broadly expressed opioid receptor gene of peripheral neurons**

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Introduction

Opioid signaling in vertebrates runs through four class A G protein-coupled receptors: the mu (MOR), delta (DOR), and kappa (KOR) opioid receptors, encoded by *OPRM1*, *OPRD1*, and *OPRK1*, and the nociceptin/orphanin FQ peptide receptor (NOP), encoded by *OPRL1*. All four couple to Gi/Go and converge on a shared set of neuronal effectors: inhibition of adenylyl cyclase, activation of GIRK/Kir3 channels, and voltage-dependent suppression of Cav2.1 and Cav2.2 currents, which lowers somatic excitability and reduces transmitter release at nerve terminals (Al-Hasani and Bruchas, 2011). Selectivity arises at the orthosteric pocket. Beta-endorphin and the enkephalins act at MOR and DOR, dynorphins at KOR, and nociceptin/orphanin FQ (N/OFQ) at NOP.

NOP was cloned as an orphan receptor sharing approximately 47% amino acid identity with MOR, DOR, and KOR overall and 64% to 67% identity across the transmembrane helices (Mollereau et al., 1994; Bunzow et al., 1994). Its endogenous ligand was identified a year later (Meunier et al., 1995; Reinscheid et al., 1995). N/OFQ derives from prepronociceptin (*PNOC*) and begins with Phe-Gly-Gly-Phe in place of the Tyr-Gly-Gly-Phe motif shared by the classical opioid peptides (Mollereau et al., 1996). This substitution abolishes cross-reactivity in both directions, and naloxone and the morphinan alkaloids have negligible affinity for NOP. The antagonist-bound crystal structure accounts for the pharmacological separation: a glutamine occupies the position of the transmembrane helix 6 histidine conserved across the classical receptors, and acidic residues in extracellular loop 2 reshape the entrance to the pocket (Thompson et al., 2012). NOP therefore shares an effector repertoire with MOR, DOR, and KOR while occupying a distinct ligand space.

Peripheral opioid receptors support analgesia and visceral regulation independently of the brain. Leukocytes recruited to inflamed tissue synthesize and release POMC- and PENK-derived peptides that act on receptors at sensory terminals (Stein and Machelska, 2011). Deleting MOR or DOR selectively from primary afferents reduces opioid analgesia in inflammatory and neuropathic models (Gaveriaux-Ruff et al., 2011; Weibel et al., 2013). Drugs built on peripheral restriction are in clinical use: loperamide and the peripherally acting MOR antagonists methylnaltrexone, naloxegol, naldemedine, and alvimopan act on enteric MOR, and difelikefalin, a peripherally restricted KOR agonist, was approved in 2021 for pruritus in patients on hemodialysis (Fishbane et al., 2020). Every approved peripheral opioid drug targets MOR or KOR.

Transcriptomic surveys of sensory ganglia have sharpened this picture and exposed species differences. In human dorsal root ganglion (DRG), *OPRM1* is enriched across peptidergic nociceptor classes and *OPRD1* marks non-peptidergic and itch-associated populations, while mouse sensory neurons show no comparable cell-type partitioning (Tavares-Ferreira et al., 2022; Bhuiyan et al., 2024). *OPRK1* transcripts sit near the detection floor in human DRG, although KOR protein and KOR-mediated inhibition of voltage-gated calcium currents are demonstrable in the same tissue (Snyder et al., 2018). Evidence for peripheral NOP has accumulated separately from these atlases. NOP immunoreactivity is present in 75% to 80% of small and medium human DRG neurons and rises several-fold in suburothelial fibers from patients with detrusor overactivity and bladder pain syndrome (Anand et al., 2016). N/OFQ inhibits Cav2.2 currents in dorsal root, superior cervical, and vestibular ganglion neurons, suppresses T-type currents through a G protein-independent route (Abdulla and Smith, 1997), modulates N-type channels tonically in the absence of agonist (Beedle et al., 2004), and reduces capsaicin-evoked responses

in cultured human sensory neurons at picomolar concentrations (Anand et al., 2016). NOP and N/OFQ are also present in enteric neurons, where they alter motility and secretion (Drokhlyansky et al., 2020; Toll et al., 2016). The abundance of the four receptors relative to one another across peripheral neuron types has not been measured on a common scale, and NOP has been characterized mainly as a central receptor with anti-opioid actions at supraspinal sites.

We find that *Oprl1* is the most broadly expressed opioid receptor gene in neurons of the peripheral nervous system. It carries the highest population mean of the four in 18 of 19 neuronal populations across the sensory, sympathetic, parasympathetic, and enteric divisions, and it is detected in more neurons than *Oprm1*, *Oprd1*, or *Oprk1* in almost every population in which all four detection rates are recorded. Among the neurons that do express them the four sit far closer together, so the difference is principally one of prevalence rather than of abundance per neuron. The distribution we measure inverts the rank order assumed by current peripheral opioid pharmacology, and identifies *Oprl1* as the opioid receptor transcript most widely available to peripheral neurons. Whether NOP protein follows the transcript is not established by these data.

Results

***Oprl1* is the most broadly expressed opioid receptor gene across the peripheral nervous system**

Every peripherally restricted opioid drug in clinical use acts at MOR or KOR. The four opioid receptors have rarely been measured against one another in the same peripheral neurons. We ranked *Oprl1*, *Oprm1*, *Oprd1*, and *Oprk1* within each of 19 neuronal populations from 16 mouse ganglia and plexuses, covering the sensory, sympathetic, parasympathetic, and enteric divisions. The data come from 12 published single-cell studies deposited between 2016 and 2026; we generated no new sequencing (Table S1). We took the whole-cell fraction of each deposit and applied one analysis to all of them. *Oprl1* carries the highest population mean of the four in 18 of the 19 populations.

The cell bootstrap reported throughout describes how far a population's ordering depends on which cells were captured. It is not a test across animals, and four of the nineteen populations come from a single one, so the inference is made at the unit the claim is about. Taking the population as that unit, *Oprl1* leads 18 of 19 (sign test against a coin-flip null, $p = 3.8 \times 10^{-5}$). Taking the deposit, so that the five autonomic ganglia of one experiment count once and a deposit leads only if *Oprl1* is first in every population it contributes, 11 of 12 ($p = 0.0032$). The one deposit that does not lead is GSE263422, on the strength of its postnatal day 7 enteric sample.

The ranking holds in SMART-seq full-length libraries sequenced to a median of 2.7 million reads per neuron and in droplet libraries at 1,570 UMI per cell. Separation from the other three receptors is widest in the spiral ganglion, where *Oprl1* reaches 41.94 CPM in 74.8% of neurons and *Oprm1*, *Oprd1*, and *Oprk1* are absent from all 226 cells, and in the sphenopalatine ganglion, where *Oprl1* reaches 19.50 CPM against *Oprm1* at 0.29, *Oprd1* at 0.04, and *Oprk1* at 0.01 CPM (Figures 1A and 1P). The two largest droplet datasets give the narrowest separations: *Oprl1* exceeds *Oprm1* by 1.21-fold in 26,047 nodose neurons and by 1.13-fold in 31,802 dorsal root ganglion neurons, each holding in every bootstrap replicate with a 95% confidence interval excluding 1 (Figures 1F and 1H).

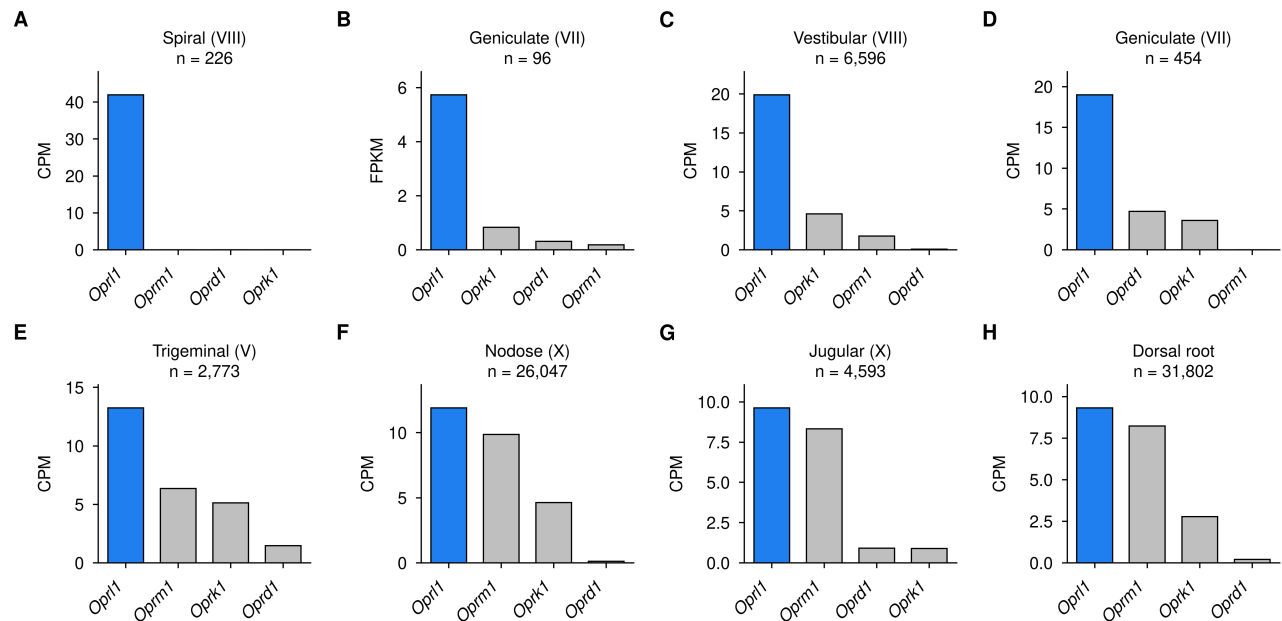
The geniculate ganglion was sequenced full-length by Dvoryanchikov et al. (2017) and by droplet capture by Zhang et al. (2019), and the stellate ganglion appears in two unrelated studies. *Oprl1* ranks first in all four (Figures 1B, 1D, 1N, and 1O).

Five populations were dissected and sequenced together in one experiment (Sivori et al., 2024), which places them on a common scale. Across four sympathetic ganglia and the sphenopalatine ganglion, *Oprl1* varies 1.7-fold, from 16.33 CPM in the pelvic ganglion to 27.55 CPM in the stellate ganglion. The other three receptors vary between 6-fold and 302-fold over the same five populations. Across the full panel, *Oprm1*, *Oprk1*, and *Oprd1* each rank second in some populations, and none of them ranks second consistently. NOP is available to autonomic neurons at a similar level regardless of ganglion, and MOR, DOR, and KOR are distributed by division and cell type.

The enteric submucosal plexus before weaning ranks *Oprk1* first. At postnatal day 7, *Oprk1* reaches 122.51 CPM against 26.70 CPM for *Oprl1* in both samples at that age (Figure 1S). By postnatal day 24, *Oprk1* has fallen to 3.77 CPM at 1.1% detection, *Oprl1* stands at 35.57 CPM, and *Oprl1* ranks first in all three samples (Figure 1R). Both ages come from one deposit, one tissue, and one platform (Li et al., 2025), and the 32-fold fall in *Oprk1* is the largest change of the four receptors between them: *Oprd1* rises 7-fold, *Oprm1* falls 4-fold, and *Oprl1* moves 1.3-fold. High submucosal *Oprk1* is a property of the neonatal plexus, and the mature plexus matches the rest of the panel.

The difference between *Oprl1* and the other three is principally one of prevalence. In the 15 populations that record detection for all four receptors, *Oprl1* is the most widely detected in 14, reaching 7.2% to 91.7% of neurons against 0.2% to 12.0% for *Oprm1* (Figure S1A). Among the neurons that do express them the receptors sit much closer together: the median margin falls from 3.97-fold on the population mean to 1.17-fold on the mean among positive cells, and in 2 of the 8 populations where a second receptor is detected in at least 100 neurons the ordering changes, the vestibular ganglion and the jugular (Figures S1B and S1C). A mean over positive cells is floored by the one-count detection limit and therefore favours the rarely detected receptors, so that comparison is conservative for prevalence and unreliable where the positive-cell count is small. *Oprl1* is available to more peripheral neurons than the other three opioid receptors rather than present at a higher level in the neurons that carry them.

Sensory



Parasympathetic and enteric

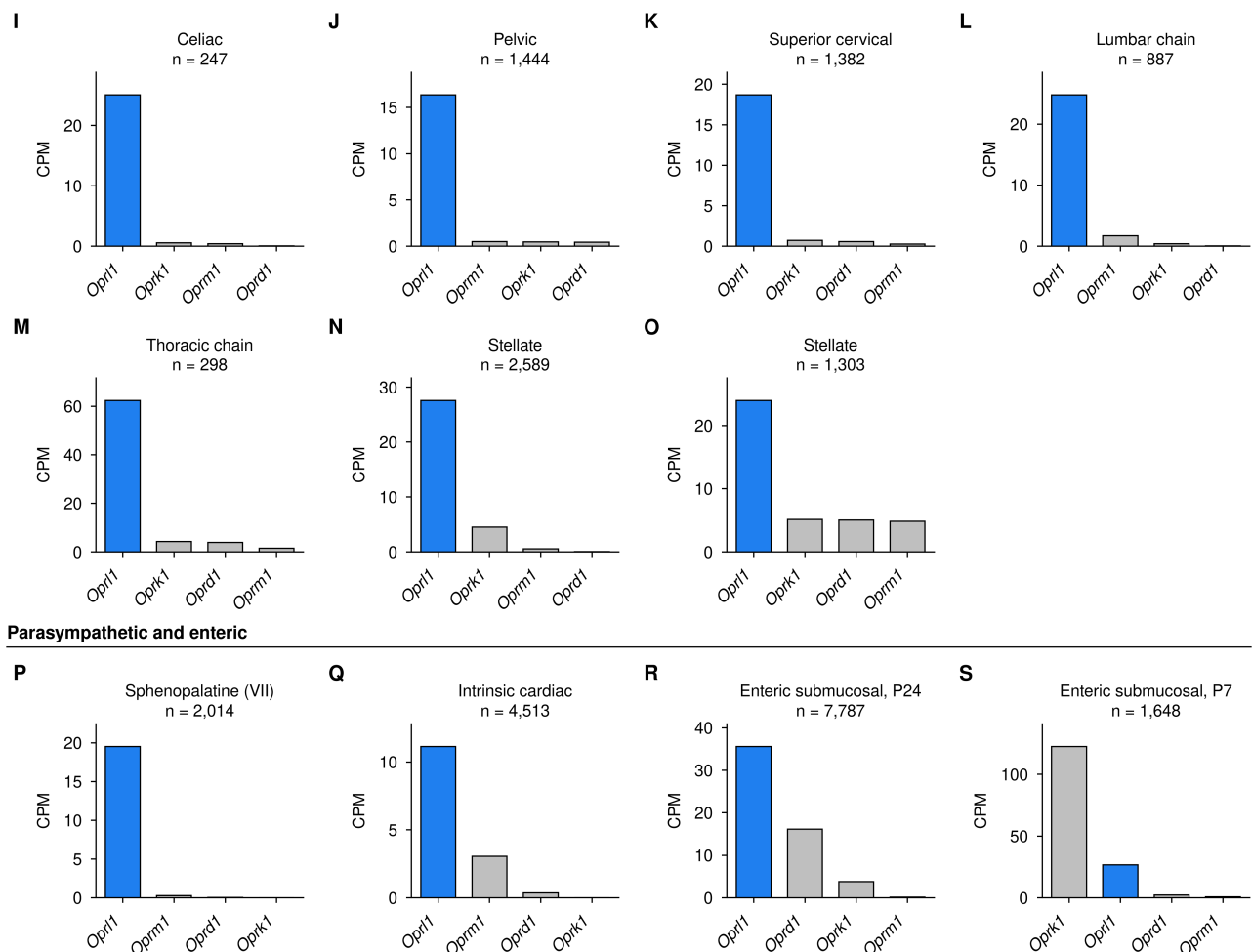


Figure 1. *Oprl1* leads the four opioid receptor genes in 18 of 19 peripheral populations

Mean expression of the four opioid receptor genes in each of 19 peripheral neuronal populations, grouped by division of the peripheral nervous system. Roman numerals give the cranial nerve of ganglia that belong to one. Bars are pseudobulk means over all neurons of the population, in the unit each dataset was measured in, ordered within each panel from highest to lowest. *Oprl1* is blue and the other three receptors are gray. Sample size is the number of neurons contributing to the panel.

(A) Spiral ganglion (VIII), GSE114997, full-length SMART-seq, median 2,678,701 reads per neuron. *Oprl1* 41.94 CPM in 74.8% of neurons; *Oprm1*, *Oprd1*, and *Oprk1* detected in no cell. All four genes are present in the annotation, so the three zeros are measured absences.

(B and D) Geniculate ganglion (VII) on two platforms from two laboratories. (B) GSE102443, full-length SMART-seq, FPKM, 6.92-fold over the second receptor (4.26 to 13.32), support 1.000, and 31.30-fold over *Oprm1*, which is the lowest of the four here. (D) GSE135801, 3' droplet, 4.02-fold, support 0.974 (95% interval 0.99 to 36.70).

(C) Vestibular ganglion (VIII), GSE309608, four mice, 4.29-fold (3.85 to 4.81), support 1.000, same ordering in each mouse.

(E) Trigeminal ganglion (V), iPain atlas, whole-cell neurons only, 2.08-fold (1.41 to 3.26).

(F and G) Inferior (nodose) and superior (jugular) ganglia of the vagus (X), NodoseMap atlas, whole-cell datasets only. Nodose 1.21-fold (1.15 to 1.28), support 1.000, first in all four contributing datasets; jugular 1.16-fold (0.92 to 1.46), support 0.898, first in two of three.

(H) Dorsal root ganglion, a spinal ganglion with no cranial nerve, iPain atlas, whole-cell neurons only, 1.13-fold (1.08 to 1.19), the narrowest margin measured and the second largest sample.

(I to O) Sympathetic ganglia, ordered by margin. Celiac 44.41-fold (20.42 to 197.63), pelvic 32.42-fold (22.00 to 54.20), superior cervical 25.72-fold (15.93 to 52.14), lumbar chain 14.78-fold (12.00 to 18.83), thoracic chain 14.69-fold (9.35 to 27.27), stellate 6.08-fold (5.16 to 7.27), stellate 4.68-fold (3.02 to 8.08). Support 1.000 throughout. (I), (J), (L), and (N) are from GSE232789; (K) from the two untreated animals of GSE231766; (M) from GSE78845; (O) from GSE231924, which profiles cardiac-projecting neurons and places *Oprl1* first in each of eight mice. The stellate ganglion appears twice, in (N) and (O), from two deposits, two laboratories, and two neuron-calling routes. The pelvic ganglion carries both sympathetic and parasympathetic neurons and is grouped here with the sympathetic block.

(P and Q) Parasympathetic ganglia. (P) Sphenopalatine ganglion (VII), also called the pterygopalatine ganglion, GSE232789, 66.86-fold (40.54 to 132.54), the largest margin in the study. These neurons are cholinergic and not noradrenergic (*Slc18a3* 252 CPM, *Chat* 26, *Th* 2, *Dbh* 42) against *Th* 390 to 840 CPM in the sympathetic ganglia of the same experiment. (Q) Intrinsic cardiac nervous system, GSE330884, three mice, 3.65-fold (3.29 to 4.08), the only population in this division in which *Oprm1* is the second receptor. These ganglia are intrinsic to the heart and receive vagal preganglionic input, so they carry no cranial nerve numeral.

(R and S) Enteric submucosal neurons of the small intestine at two ages, GSE263422, one laboratory and one platform. (R) Postnatal day 24, *Oprl1* first at 2.20-fold (2.05 to 2.37) in all three samples. (S) Postnatal day 7, *Oprk1* first at 122.51 CPM against *Oprl1* 26.70, 4.59-fold (4.04 to 5.24) in both samples. *Oprk1* falls to 3.77 CPM at 1.1% detection by day 24 while *Oprl1* holds between 26 and 36 CPM.

All panels use whole-cell data. Both iPain atlases are majority single-nucleus and are restricted here to `suspension_type == "cell"`; pooling preparations compresses the trigeminal margin from 2.08-fold to 1.18-fold, for the reason given in Figure 2. Neurons were called on raw counts with glial and immune barcodes excluded, and every population passed the same marker gate and ambient-RNA check before any receptor value was read from it. Margins are the ratio of the highest to the second-highest receptor; intervals in parentheses are 95% bootstrap intervals over 10,000 resamples of the cells, and support is the fraction of those resamples retaining the observed top receptor. The bootstrap resamples cells and therefore describes how far a population's ordering depends on which cells were captured; it is not an interval across animals, and the inference across populations and deposits is the sign test given in the Results. Panels (I), (J), (L), (N), and (P) come from one study of five autonomic ganglia in six samples, so laboratory and platform are constant across those five. That deposit was sequenced in two batches, stellate with sphenopalatine and the first pelvic sample, lumbar chain with celiac and the second, and median library ranges 2.6-fold across the five, from 27,931 to 73,326 UMI.

See also Figure 2, Figure S1, Figure S2, and Table S1.

Nuclear libraries scale with gene length and whole-cell libraries do not

Single-nucleus libraries from vagal and dorsal root ganglia place *Oprm1* first, at 244.49 CPM against 5.22 for *Oprl1* and at 94.92 against 7.60, and the whole-cell fractions of those same atlases place *Oprl1* first (Figures 2A and 2B). The two preparations sample different RNA. A nucleus holds transcript that is still being made, unspliced and not yet exported, so most of a nuclear library is intronic; single-nucleus quantification counts reads across the entire gene body because counting exons alone would discard that majority. The number of reads assigned to a gene then scales with the length of its transcription unit as well as with the number of transcripts the neuron has made. Whole-cell libraries sample the cytoplasmic pool, which is the spliced mRNA available for translation, and 3' counting of that pool carries no length term. Across 14,876 protein-coding genes quantified in both preparations, nuclear expression rises with genomic span ($r = +0.403$) and whole-cell expression stays flat over a 68-fold range of length ($r = +0.022$; Figure 2E). *Oprm1* spans 279.7 kb and *Oprl1* spans 7.1 kb, giving nuclear-to-

whole-cell ratios of 25.4 and 0.45 (Figure 2F), and *Oprm1* is detected in 71.9% of nuclei against 5.5% to 32.4% of cells (Figures 2C and 2D). The nuclear values report a combination of transcript abundance and gene length.

Dissociation is a second preparation effect, and it is separable from this one. Whole-cell libraries are made from live enzymatic dissociation, which axotomises every neuron, and they carry that injury: *Fos* reads 448.82 CPM in whole cells against 0.87 in nuclei, and *Atf3*, the axotomy marker, 151.72 against 6.15. Scored against genes of the same length, the twelve dissociation- induced genes sit at a median 1.9th percentile of their own length class, and 11 of the 12 fall below the lowest of the four receptors. The receptors sit at the 31.8th, 60.1st, 93.6th and 99.2nd percentiles (Figure S2). No claim here may assume the whole-cell libraries are unperturbed, and the length relationship is not a restatement of the injury response.

Several of the largest sensory ganglion atlases are single-nucleus, which is consistent with, and may have reinforced, the prevailing view that *Oprm1* is the dominant opioid receptor of peripheral neurons. That view predates those atlases and rests on pharmacology, conditional knockouts and reporter lines rather than on transcript counts.

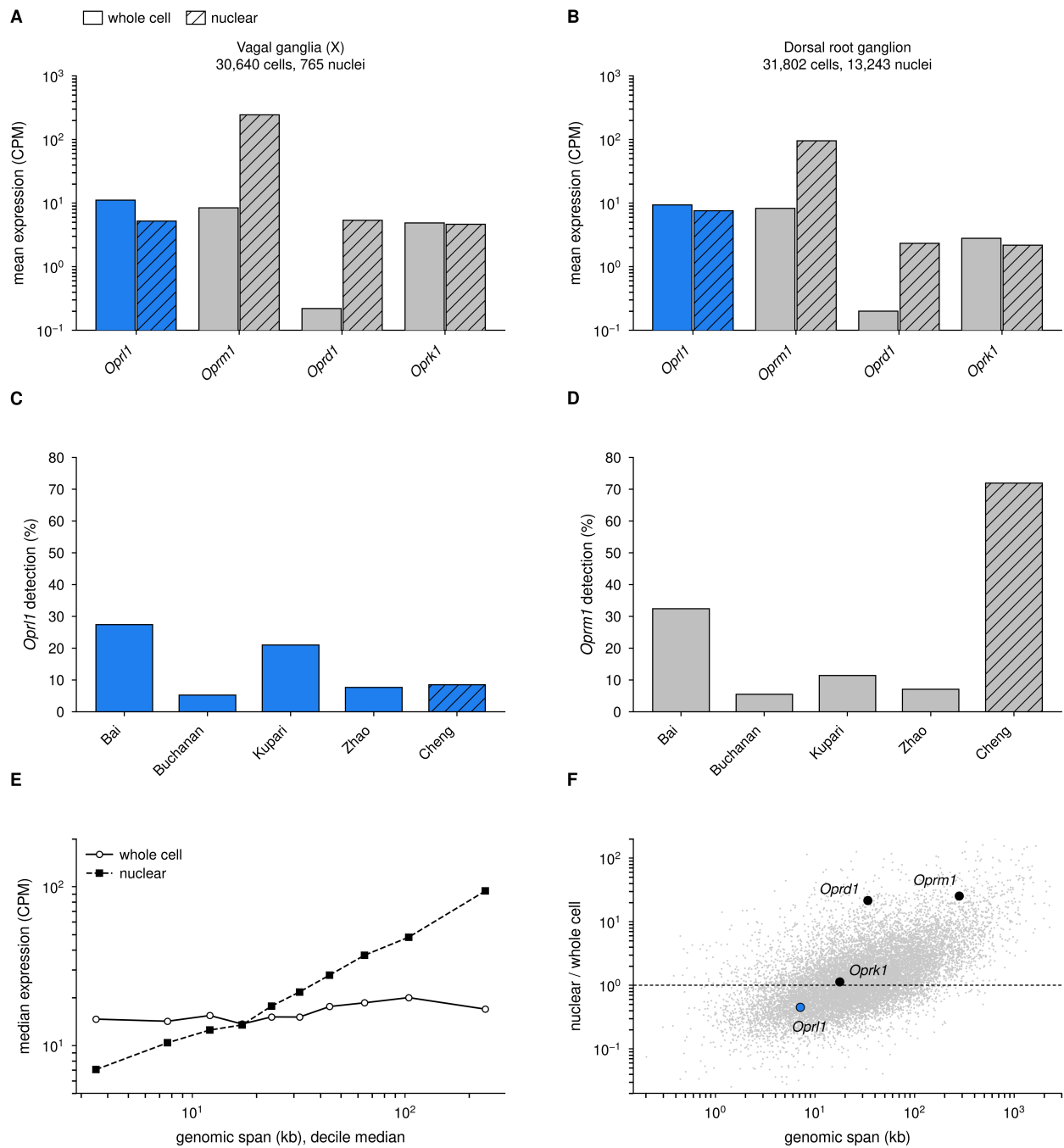


Figure 2. A nuclear preparation reverses the receptor ordering

Whole-cell and single-nucleus measurements of the same tissue, and the gene property that separates them. Solid bars are whole-cell data and hatched bars are nuclear data throughout, as the key above the panels gives. *Oprl1* is blue where the panel distinguishes genes by colour, as in Figure 1.

(A) Vagal ganglia (X), NodoMap. The whole-cell value is the unweighted mean over the atlas's four whole-cell deposits, 30,640 neurons; the nuclear value is the atlas's own 765 nuclei, the same tissue and the same integration. Whole cell: *Oprl1* 11.13 CPM, *Oprm1* 8.42, *Oprk1* 4.85, *Oprd1* 0.22. Nuclear: *Oprm1* 244.49, *Oprd1* 5.38, *Oprl1* 5.22, *Oprk1* 4.63. *Oprm1* rises 29-fold and *Oprl1* falls by half, which moves *Oprm1* from second to first.

(B) Dorsal root ganglion, a second tissue and a second laboratory. Whole cell, iPain atlas, 31,802 neurons: *Oprl1* 9.31 CPM, *Oprm1* 8.23, *Oprk1* 2.78, *Oprd1* 0.20. Nuclear, GSE201654 mouse arm, six samples and 13,243 nuclei: *Oprm1* 94.92, *Oprl1* 7.60, *Oprd1* 2.33, *Oprk1* 2.17. *Oprm1* is first in each of the six nuclear samples.

(C and D) Detection rate of *Oprl1* and of *Oprm1* in each of the five deposits that make up the NodoMap integration, on one scale. Deposits are named for the authors of the data. Bai, Buchanan, Kupari and Zhao are published cell suspensions that the NodoMap authors re-analysed; Cheng is the deposit the atlas labels in-house, the 765 nuclei the NodoMap authors generated themselves, and is the nuclear arm

of (A) and (E) as well. (C) *Oprl1* is detected in 8.5% of the nuclei, inside the 5.3% to 27.5% range of the four whole-cell deposits. (D) *Oprm1* is detected in 71.9% of the same nuclei, against 5.5% to 32.4% in the whole-cell deposits, so it exceeds the highest whole-cell value by 2.2-fold. Detection rate scales with sequencing depth, which differs between deposits; (C) is what shows that depth does not account for (D), since the two genes were counted in the same libraries.

(E) Median expression against genomic span, over the 14,876 protein-coding genes of the atlas that reach 0.1 CPM in both preparations, binned into span deciles. One marker per decile, open circles for whole cell and filled squares for nuclear, joined to show the direction of each series. Genes were selected on expression and never on length, and the floor is low enough to admit all four opioid receptors. Whole-cell medians run between 13.7 and 20.0 CPM across deciles whose median spans run from 3.5 kb to 238.4 kb, a 68-fold range of length, and correlate with span at $r = +0.022$ over the individual genes. Nuclear medians rise from 7.1 to 94.1 CPM over the same deciles, $r = +0.403$. Gene length carries no information about how much mature transcript a neuron holds, so the preparation whose levels track it is the distorted one. The expression floor is a judgement call and does not carry the result: from no floor at all to 1 CPM, the whole-cell correlation runs from +0.117 to -0.065 and the nuclear one from +0.399 to +0.433 (`results/preparation_bias_floor_sensitivity.csv`).

(F) Nuclear level divided by whole-cell level, for each of those genes, against genomic span. Each grey point is one gene. The dashed line is no change, and the four opioid receptors are marked. *Oprl1* at 7.1 kb sits at 0.45, the 32nd percentile of the 2,431 genes within a factor of 1.6 of its length, and *Oprk1* at 17.7 kb sits at 1.14 and the 60th percentile of its own length class, so both behave as ordinary genes of their size and the ordering between them survives the preparation. *Oprm1* at 279.7 kb sits at 25.4 against a median of 4.94 for genes of comparable length, the 94th percentile of that class. *Oprd1* at 33.8 kb sits at 21.6 against a median of 1.24, the 99th percentile and the largest departure of the four; its whole-cell level is 0.25 CPM, so that ratio is measured on a small denominator and is the least certain of the four. For *Oprm1* and *Oprd1* length accounts for part of the gain and not all of it. Across the deciles of (E) the median ratio runs from 0.46 to 5.22 and crosses 1 at 15.8 kb. Both panels use the vagal neurons of (A), one atlas and one chemistry, so the contrast is between preparations rather than between studies. Spans are Ensembl GRCm39 gene loci.

A nucleus holds nascent transcript that has not been spliced or exported, and single-nucleus quantification counts reads across the whole gene body, introns included, since restricting to exons discards most of a nuclear library. Signal accrues with the length of the transcription unit rather than with the number of finished transcripts. Whole-cell libraries sample the cytoplasmic pool, the mature mRNA a neuron translates, and 3' counting of that pool has no length term. This is a bias in a known direction rather than a failure of the assay, and it reaches the opioid receptors because they span 7 to 280 kb.

Preparation is confounded with laboratory in (A) and (B), since no laboratory here has run both preparations on one tissue. (E) and (F) do not rest on that contrast: both are internal to a single atlas and a single chemistry, and they identify the biased measurement from the behaviour of 14,876 genes rather than from any receptor.

Every panel of Figure 1 therefore uses whole-cell data, and no human single-nucleus dataset can establish a species difference in the ordering: the mouse arm of the one cross-species nuclear experiment, shown in (B), reverses in the same direction.

Discussion

Single-nucleus and whole-cell preparations of the same ganglia disagree about which opioid receptor predominates, and the disagreement follows gene length. The two preparations sample different RNA and capture it by different chemistry. A nucleus holds transcript that is still being made, unspliced and not yet exported, so most of a nuclear library is pre-mRNA; quantification counts reads across the entire gene body, because restricting to exons would discard that majority. Oligo(dT) primers hybridize to internal adenosine homopolymers as well as to the poly(A) tail. These homopolymers occur frequently in introns, and their number rises with the length of the transcription unit, so reads assigned to a gene scale with its genomic span as well as with the number of transcripts the neuron has made (Chamberlin et al., 2024). Mature mRNA carries a single priming site at its 3' end, and whole-cell libraries therefore record abundance without reference to length. Across 14,876 protein-coding genes, nuclear expression rises with genomic span ($r = +0.403$) while whole-cell expression stays flat over a 68-fold range ($r = +0.022$), and the four receptors order by span: *Oprm1* (279.7 kb) is enriched 25.4-fold in nuclei, *Oprd1* (33.8 kb) and *Oprk1* (17.7 kb) fall between, and *Oprl1* (7.1 kb) is depleted to 0.45. Span accounts for a minority of the variance in this ratio, and *Oprm1* lies above the trend for genes of comparable length. Gene length indexes pre-mRNA content imperfectly, and loci with long first introns, dual promoters and slow splicing, a description that fits *Oprm1* (Liu et al., 2021), accumulate nuclear signal beyond what span predicts; global length scaling will over-correct rapidly spliced genes and under-correct *Oprm1*. Dissociation offers a competing account, since it

axotomizes every neuron and axotomy lowers opioid responsiveness while raising nociceptin responsiveness in DRG neurons (Abdulla and Smith, 1998). That account is not idle here: the whole-cell libraries carry the injury programme plainly, with *Fos* at 448.82 CPM against 0.87 in nuclei and *Atf3* at 151.72 against 6.15, and the twelve dissociation-induced genes sit at a median 1.9th percentile of their own length class. The two effects are nonetheless separable, because the receptors sit at the 31.8th to 99.2nd percentiles of theirs and 11 of the 12 induced genes fall below the lowest of them (Figure S2). Dissociation perturbs these libraries and does not account for the reversal; what it leaves open is a smaller contribution to the receptors themselves, which only measurement in undissociated tissue can close. Ranking genes by CPM within a nuclear library confounds transcript abundance with transcription unit length. MOR primacy in peripheral analgesia rests on knockout phenotypes, reporter knock-in lines, radioligand binding and in situ hybridization (Scherrer et al., 2009; Wang et al., 2018); single-nucleus atlases have inflated that hierarchy rather than established it.

The transcript measurement reported here is not itself evidence about receptor protein, and the orthogonal evidence for peripheral NOP was gathered before these atlases existed. NOP immunoreactivity is present in 75% to 80% of small and medium human dorsal root ganglion neurons and rises in suburothelial fibers in bladder pain syndrome (Anand et al., 2016), a prevalence of the same order as the transcript prevalence measured here. N/OFQ inhibits Cav2.2 currents in dorsal root, superior cervical, and vestibular ganglion neurons (Abdulla and Smith, 1997; Beedle et al., 2004), which are three of the ganglia in this survey, modulates N-type channels tonically in the absence of agonist (Beedle et al., 2004), and reduces capsaicin-evoked responses in cultured human sensory neurons at picomolar concentrations (Anand et al., 2016). Protein and function therefore converge on the distribution the transcripts describe. What no dataset here establishes is that relative transcript abundance predicts relative receptor density, and the claim this work supports is about the availability of the transcript rather than about a pharmacological hierarchy.

Methods

Data

No new sequencing was generated. All data are public single-cell or single-nucleus RNA-seq deposits, listed with their accessions, studies, platforms, sample sizes and quality-control outcomes in Table S1. Nineteen neuronal populations from 16 mouse ganglia and plexuses were drawn from 12 deposits published between 2016 and 2026, and were reached through GEO and CZ CELLxGENE.

Every value reported in Figure 1 comes from whole-cell libraries. Both iPain atlases and the NodoMap atlas contain nuclear as well as whole-cell fractions; those were restricted to `suspension_type == "cell"` before any receptor value was read, for the reason established in Figure 2. The nuclear fractions are used only in Figure 2, where they are the subject.

Quantification

Expression is per-cell counts per million, computed as the count of a gene in a cell divided by that cell's total counts and multiplied by 10^6 , and a population's level is the mean of that quantity over all its neurons. GSE102443 was deposited as FPKM and is analysed in that unit; levels are compared only within a population, never across units or platforms. GSE231924 was deposited as Seurat log-normalised values at a scale of 10,000, from which CPM is recovered as $\text{expm1}(x) \times 100$, with the scale verified from the row sums rather than assumed.

A gene absent from a dataset's annotation is recorded as a null and never as a measured zero. Every level table carries an `in_matrix` field derived from the matrix itself, so that a reference gap and a measured absence remain

distinguishable. The three zeros in the spiral ganglion are measured absences by that test: all four receptors are present in the annotation and three are detected in none of the 226 neurons.

Cell filtering and neuron calling

Barcodes were retained at 2,000 or more UMI and 1,000 or more detected genes. Neurons were then called on raw counts, so that the threshold does not move with library size, and glial and immune barcodes were excluded on CPM, so that those thresholds do not move with sequencing run.

The neuronal criterion is *Snap25* at 5 counts or more together with a second marker chosen for the tissue: *Tubb3* for the autonomic and vestibular ganglia, *Th* for the superior cervical ganglion, *Phox2b* for the intrinsic cardiac nervous system, and *Elavl4* for the enteric plexus. Barcodes carrying a glial, myelinating or immune signature were excluded whatever their *Snap25* count, at *Sox10* below 200 CPM, *Plp1* below 500 CPM and *Ptprc* below 50 CPM. *Snap25* alone admits ambient-dominated barcodes, which is what these exclusions remove. Populations deposited as sorted or author-filtered neurons were taken as supplied.

Marker gate

Before any receptor number was read from a population, its identity was confirmed against *Snap25*, *Phox2b*, *Slc17a6*, *Tac1*, *Calca* and *Actb*. *Snap25* and *Actb* are enforced: they must be present in the matrix and non-zero, and the pipeline raises rather than warns if they are not. The remaining markers are tissue-specific, and are recorded without being enforced. All 21 populations in the quality panel pass.

Ambient RNA

No ambient correction is applied anywhere in this work, so a transcript read in neurons carries whatever the same transcript contributes to the surrounding soup. That contribution is measured instead of removed: within a single dissociation, each receptor's mean CPM in the called neurons is compared with its mean CPM in the non-neuronal barcodes captured beside them, as a log₂ ratio with a pseudocount of 0.01 CPM. *Plp1* and *Ptprc* serve as negative controls.

Oprl1's enrichment on its own conflates how neuronal the transcript is with how cleanly the two compartments separated in that dissociation. *Snap25* is neuronal by definition and calibrates the second, so the quantity reported is *Oprl1*'s enrichment minus *Snap25*'s. Across the eleven peripheral populations where both are available, *Oprl1* sits within 0.67 log₂ of *Snap25*, median -0.14. The check ran on 14 of 21 populations; a deposit of sorted or author-filtered neurons has no non-neuronal compartment to score against, and the reason is recorded per population rather than left as a gap.

Receptor ordering, bootstrap and sample agreement

The four receptors are ranked within a population, on the same cells and the same chemistry, which is what makes the ordering comparable between populations measured on different platforms. A rank is reported as determinate only when the top receptor is strictly above the runner-up. The margin is the ratio of the highest receptor to the second-highest.

Uncertainty in that ordering is estimated by resampling the cells of a population with replacement, 10,000 times, from a fixed seed. The 95% interval on the margin is the 2.5th and 97.5th percentile of the resampled margins, and support is the fraction of resamples in which the observed top receptor is still top. A pseudobulk ordering with a 1.8% margin and one with a 27% margin are otherwise reported identically, which is what this number prevents.

Where a deposit resolves into biological samples, each sample was also ordered on its own cells, and the count of samples placing the top receptor first is reported alongside the bootstrap.

The bootstrap resamples cells, so it describes one population and not a difference between animals. Cells within a ganglion are not independent replicates, four of the nineteen populations come from a single animal, and nine do not record how many contributed, so no interval computed over cells is treated as inference about the claim the atlas makes. That claim is tested at its own unit with a sign test: the count of populations, and more conservatively of deposits, that place *Oprl1* first, against a null in which the four receptors are exchangeable ($p = 0.25$) and against the more conservative null of *Oprl1* against the field ($p = 0.5$). A deposit counts as leading only if *Oprl1* is first in every population it contributes, so the five autonomic ganglia of one experiment enter once and GSE263422 is scored against the atlas on its postnatal day 7 sample.

Prevalence against level in positive cells

A population mean is the product of the fraction of neurons in which a gene is detected and its mean level in those neurons, and the two are reported separately wherever both are recorded, which is 15 of the 19 populations. The mean among positive cells is the population mean divided by the detection fraction.

That quantity is biased, in a direction that matters. A droplet library cannot record less than one count, so a gene detected in few cells has its conditional mean floored by the quantisation limit rather than measured, which inflates the rarely detected receptors and works against *Oprl1*. The comparison is therefore conservative for the prevalence claim, and no ordering is read from a gene detected in fewer than 100 neurons. The count of positive cells is reported for every entry.

Whole-cell against nuclear preparation

Figure 2 compares the two preparations within the NodoMap integration: the same tissue, the same chemistry and the same annotation, so the contrast is between preparations rather than between studies. Preparation is nonetheless confounded with laboratory in the two paired tissue comparisons, since no laboratory here ran both preparations on one tissue, and the genome-wide analysis is what does not rest on that contrast.

Genomic spans are Ensembl GRCm39 gene loci, fetched once by symbol lookup so that every gene is measured on one annotation. Per-dataset feature tables carry coordinates for the features they quantify, but not for every gene and not on one assembly, which had previously dropped genes from the comparison and misplaced others.

Genes were kept if they are protein-coding by Ensembl biotype and reach 0.1 CPM in both preparations, which leaves 14,876. The floor exists only to keep ratios off a near-zero denominator; selection is on expression and never on length, and it is low enough to admit all four opioid receptors, which must not be filtered out of an analysis about them. Correlations are Pearson on log10 span against log10 expression, with Spearman on the untransformed values reported alongside. The floor was varied from none at all to 1 CPM: the whole-cell correlation runs from +0.117 to -0.065 and the nuclear one from +0.399 to +0.433, so the choice of floor sharpens the separation rather than creating it. Span deciles are ten equal-count bins of genes.

Dissociation signature

The competing account of the preparation difference is dissociation rather than length: whole-cell libraries are made by live enzymatic dissociation, which axotomises every neuron, while the nuclei here come from tissue that was never dissociated alive. Twelve immediate-early and injury genes were scored exactly as the receptors are, by the ratio of nuclear to whole-cell level against the ratio of every gene within a factor of 1.6 of the same genomic span: the activity-induced set *Fos*, *Fosb*, *Jun*, *Junb*, *Jund*, *Egr1*, *Arc* and *Npas4*, the heat-shock pair *Hspa1a* and

Hspa1b, and the axotomy markers *Atf3* and *Socs3*. The gene set was fixed before the scores were computed and is not selected on the result.

Figures and tables

Figures are built to Cell Press artwork specification at the journal's fixed column widths, in Helvetica or a metric-compatible substitute, and are written at the size they will print rather than rescaled at submission. The build refuses to write a figure that exceeds the journal's width or height or that places any element outside the canvas. Table S1 is assembled from the analysis tables under `results/`; the build recomputes every margin from the levels Figure 1 is drawn from and refuses to write the table if a recorded margin disagrees with it.

Software and availability

Analysis in Python 3.11 with `pandas`, `numpy`, `scipy`, `anndata`, `h5py` and `matplotlib`. All analysis code, the intermediate tables, and the scripts that produce every figure and table in this manuscript are in the repository accompanying this paper; the accession for every dataset is in Table S1.

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Supplemental information

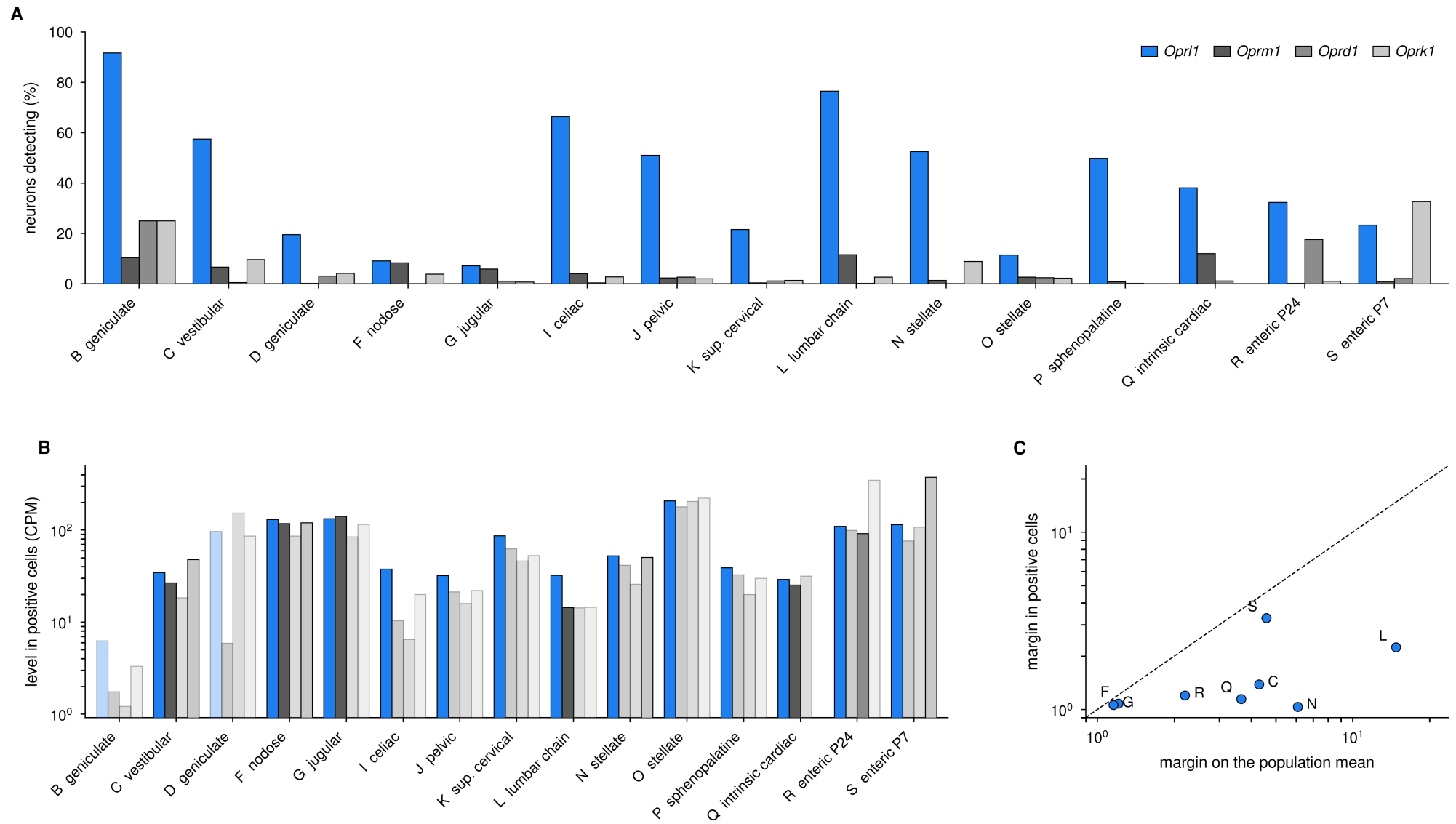


Figure S1. The lead is a difference in prevalence, related to Figure 1

A population mean is the product of the fraction of neurons in which a gene is detected and the mean level in those neurons, and the two carry different biology. Panels cover the 15 of the 19 Figure 1 populations that record detection for all four receptors; panel letters are those of Figure 1 and the tissue name follows each. *Opr1* is blue, as in Figure 1, and the other three are separated by shade rather than by hue.

(A) Percentage of neurons in which each receptor is detected. *Oprl1* is the most widely detected of the four in 14 of the 15 populations, from 7.2% in the jugular ganglion to 91.7% in the full-length geniculate deposit, against 0.2% to 12.0% for *Oprm1*. The exception is the enteric plexus at postnatal day 7, where *Oprk1* reaches 32.7% against 23.4%, the same population that reverses in Figure 1S.

(B) Mean level among the neurons that detect the gene, which is the population mean divided by the prevalence in (A). Bars are faded where fewer than 100 neurons are positive, because a mean over few positive cells is floored by the one-count detection limit rather than measured, and that floor inflates the rarely detected receptors. The four receptors sit far closer together here than in Figure 1.

(C) The margin between the top two receptors on the population mean, against the margin between the top two on the mean among positive cells, for the 8 populations where a second receptor reaches 100 positive neurons. The dashed line is equality. Every population falls below it: the median margin is 3.97-fold on the population mean and 1.17-fold among positive cells. The top receptor differs between the two measures in the vestibular ganglion (C) and the jugular (G). Because the conditional mean favours the rarely detected receptors, this panel understates the prevalence difference rather than overstating it.

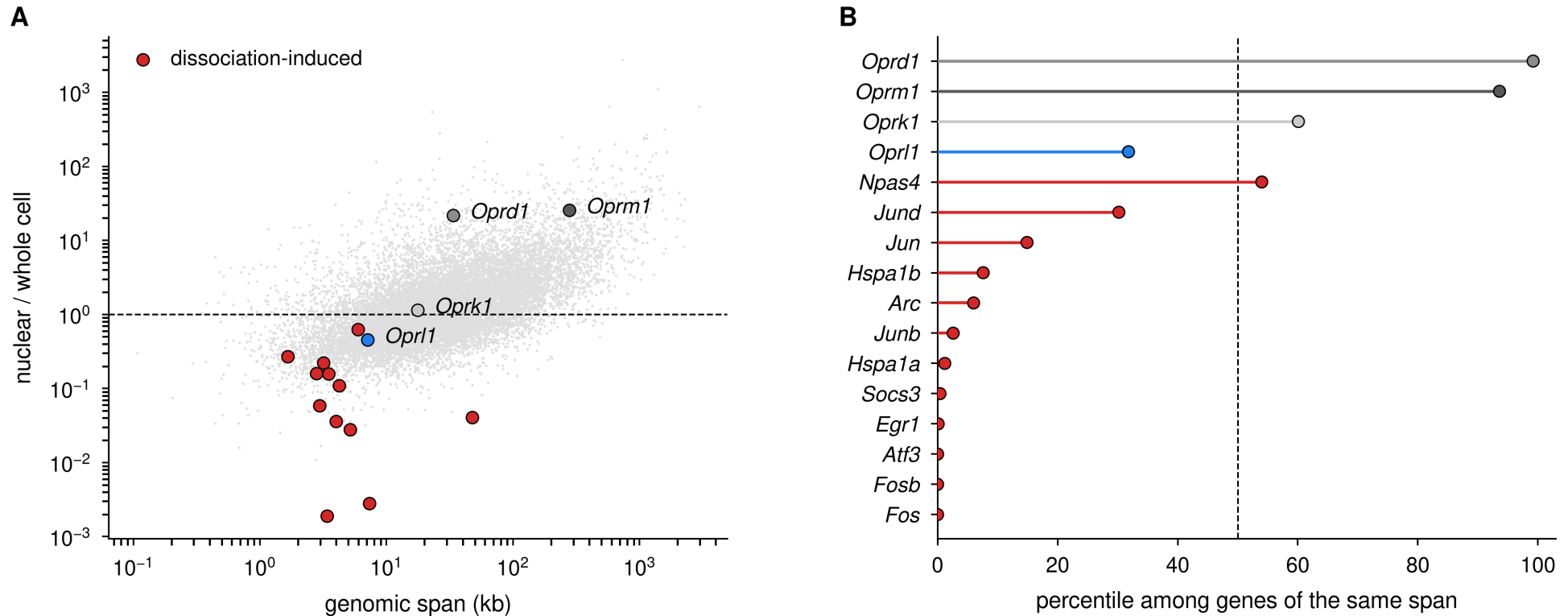


Figure S2. Dissociation is a second preparation effect and is separable from length, related to Figure 2

Whole-cell libraries are made by live enzymatic dissociation, which axotomises every neuron and induces an immediate-early and injury programme; the nuclei used in Figure 2 come from tissue that was never dissociated alive. Axotomy lowers opioid and raises nociceptin responsiveness in dorsal root ganglion neurons (Abdulla and Smith, 1998), which is the direction that would produce this atlas's result artefactually. Both panels use the 14,876 genes of Figure 2E and 2F.

(A) Nuclear level divided by whole-cell level against genomic span, as in Figure 2F. Grey points are all 14,876 genes, red points the twelve dissociation-induced genes, and the four opioid receptors are marked and coloured as in Figure S1. The dissociation genes lie at the floor of the cloud at every span; the receptors lie within it or above it.

(B) The same genes as a percentile of the ratio among genes within a factor of 1.6 of the same span, which is what separates an induced gene from a long one. The dashed line is the median of each gene's own length class. The twelve dissociation genes sit at a median 1.9th percentile, 9 of them below the 10th, and 11 of the 12 fall below the lowest of the four receptors; *Fos* reads 448.82 CPM in whole cells against 0.87 in nuclei and *Atf3* 151.72 against 6.15. The receptors sit at the 31.8th (*Oprl1*), 60.1st (*Oprk1*), 93.6th (*Oprm1*) and 99.2nd (*Oprd1*) percentiles.

The whole-cell libraries therefore carry a real and large injury programme, and no claim in this work assumes otherwise. The four receptors do not sit in it, so the length relationship of Figure 2 is not a restatement of the dissociation response. Neither analysis can exclude a small dissociation contribution to the receptors themselves, which is a question for measurement in undissociated tissue.

Table S1. The nineteen peripheral neuronal populations of Figure 1, related to Figure 1

One row per panel of Figure 1: the deposit each population was read from, the preparation and chemistry it was measured on, the neurons and biological samples contributing, the mean level of each of the four opioid receptor genes, and the statistics behind the ordering. Panel letters are those of Figure 1. The table the journal receives is `paper/tables/TableS1.csv`, one sheet of twenty-six columns; `paper/tables/TableS1.md` renders the same rows for reading, split into (a) provenance and sample and (b) levels, ordering and quality control. Both are written by `paper/tables.py` from the tables under `results/`.

Study and DOI name the publication that generated each deposit, taken from the deposit itself rather than from a search on the tissue: eleven of the twelve GEO records name their publication, either in the citation field or in the paper's own data availability statement. GSE309608 is the exception and is matched by its BioProject accession (PRJNA1335542), its six contributors and its title, all of which the PNAS paper carries. Full citations are in `paper/references.md`.

Divisions are the blocks of Figure 1. The pelvic ganglion carries both sympathetic and parasympathetic neurons and is grouped with the sympathetic block, as in the figure; `results/dataset_quality_panel.csv` records it as mixed autonomic. All nineteen populations are whole-cell, for the reason given in Figure 2; both iPain atlases are majority single-nucleus and are restricted here to `suspension_type == "cell"`.

Levels are pseudobulk means over all neurons of the population, in the unit each dataset was measured in: FPKM for the full-length geniculate deposit, CPM elsewhere. Levels are compared only within a panel, never across units or platforms. Detection is the percentage of that population's neurons in which *Oprl1* is called, and is reported per dataset for the same reason: it scales with sequencing depth, which differs by two orders of magnitude across these deposits. Median library is per neuron, in UMI for droplet deposits and reads for full-length ones.

Margins are the ratio of the highest receptor to the second-highest on the same cells. Intervals are 95% bootstrap intervals over 10,000 resamples of the cells, and support is the fraction of those resamples that retain the observed top receptor. Samples agreeing counts the biological samples that place the top receptor first on their own cells.

Neurons were called on raw counts with glial and immune barcodes excluded, and every population passed the same marker gate before any receptor value was read from it. The ambient-RNA check is the neuron-to-non-neuron ratio of each receptor within one dissociation; where it could not run, the table gives the reason rather than a blank, and those reasons are listed under the rendered table. A deposit of sorted or author-filtered neurons has no non-neuronal compartment to score against, which is what most of them are.

Blank cells are quantities the source tables do not record, and are left empty rather than carried over from a comparable population:

- **Samples** is recorded only for the populations `src/10_peripheral_ganglia.py` derived from source. For the rest, sample agreement is in the samples-agreeing column, which is why panel F reads 4/4 with no sample count beside it. `n/a` marks a deposit that does not resolve into biological samples here.
- **Median library** is not recorded for panels B, D, E, H and O. The two iPain populations were not re-derived from their source matrices, and the GSE231924 deposit carries no library sizes.
- **Detection** is not recorded for panels E and H, the two iPain populations, for the same reason.
- **Panel A** has no margin: the spiral ganglion quantifies all four receptors and detects one, so there is no second receptor to divide by. The three zeros are measured absences, not a reference gap.

(a) Provenance and sample

Panel	Division	Population	Deposit	Study	DOI	Platform	Preparation	Neurons	Samples	Median library
A	sensory	spiral (VIII)	GSE114997	Shrestha et al., 2018	10.1016/j.cell.2018.07.007	SMART-seq, full-length	whole cell	226	--	2,678,701
B	sensory	geniculate (VII)	GSE102443	Dvoryanchikov et al., 2017	10.1038/s41467-017-01095-1	SMART-seq, full-length	whole cell	96	--	--
C	sensory	vestibular (VIII)	GSE309608	Liu et al., 2026	10.1073/pnas.2530677123	10x droplet, 3'	whole cell	6,596	4	55,172
D	sensory	geniculate (VII)	GSE135801	Zhang et al., 2019	10.1016/j.cell.2019.08.031	droplet, 3'	whole cell	454	--	--
E	sensory	trigeminal (V)	iPain	Bhuiyan et al., 2024	10.1126/sciadv.adj9173	10x droplet, 3'	whole cell	2,773	--	--
F	sensory	nodose (X)	NodoMap	Cheng et al., 2026	10.1016/j.cpblue.2026.100072	10x droplet, 3'	whole cell	26,047	--	1,570
G	sensory	jugular (X)	NodoMap	Cheng et al., 2026	10.1016/j.cpblue.2026.100072	10x droplet, 3'	whole cell	4,593	--	1,570
H	sensory	dorsal root	iPain	Bhuiyan et al., 2024	10.1126/sciadv.adj9173	10x droplet, 3'	whole cell	31,802	--	--
I	sympathetic	celiac	GSE232789	Sivori et al., 2024	10.7554/eLife.91576	10x droplet, 3'	whole cell	247	1	56,023
J	sympathetic	pelvic	GSE232789	Sivori et al., 2024	10.7554/eLife.91576	10x droplet, 3'	whole cell	1,444	2	40,986
K	sympathetic	superior cervical	GSE231766	Ziegler et al., 2023	10.1126/science.abn6366	10x droplet, 3'	whole cell	1,382	2	9,885
L	sympathetic	lumbar chain	GSE232789	Sivori et al., 2024	10.7554/eLife.91576	10x droplet, 3'	whole cell	887	1	73,326
M	sympathetic	thoracic chain	GSE78845	Furlan et al., 2016	10.1038/nn.4376	full-length	whole cell	298	--	33,099
N	sympathetic	stellate	GSE232789	Sivori et al., 2024	10.7554/eLife.91576	10x droplet, 3'	whole cell	2,589	1	27,931
O	sympathetic	stellate	GSE231924	Sharma et al., 2023	10.7554/eLife.86295	10x droplet, 3'	whole cell	1,303	--	--
P	parasympathetic and enteric	sphenopalatine (VII)	GSE232789	Sivori et al., 2024	10.7554/eLife.91576	10x droplet, 3'	whole cell	2,014	1	37,416
Q	parasympathetic and enteric	intrinsic cardiac	GSE330884	Xu et al., 2026	10.1016/j.cell.2026.06.040	10x droplet, 3'	whole cell	4,513	3	52,533
R	parasympathetic and enteric	enteric submucosal, P24	GSE263422	Li et al., 2025	10.1038/s41593-025-01962-x	10x droplet, 3'	whole cell	7,787	3	12,874
S	parasympathetic and enteric	enteric submucosal, P7	GSE263422	Li et al., 2025	10.1038/s41593-025-01962-x	10x droplet, 3'	whole cell	1,648	2	8,974

(b) Levels, ordering and quality control

Panel	Population	Unit	Opr1	Oprm1	Oprd1	Oprk1	Opr1 detected (%)	Top	Runner-up	Margin	95% CI	Support	Samples agreeing	Marker gate	Ambient check
A	spiral (VIII)	CPM	41.94	0.00	0.00	0.00	74.8	Opr1	none detected	undefined	--	1.000	n/a	pass	not possible
B	geniculate (VII)	FPKM	5.73	0.18	0.31	0.83	91.7	Opr1	Oprk1	6.92x	4.26 to 13.32	1.000	n/a	pass	not possible
C	vestibular (VIII)	CPM	19.88	1.78	0.09	4.63	57.6	Opr1	Oprk1	4.29x	3.85 to 4.81	1.000	4/4	pass	run
D	geniculate (VII)	CPM	18.98	0.01	4.72	3.61	19.6	Opr1	Oprd1	4.02x	0.99 to 36.70	0.974	n/a	pass	not possible
E	trigeminal (V)	CPM	13.24	6.36	1.46	5.12	--	Opr1	Oprm1	2.08x	1.41 to 3.26	1.000	n/a	pass	not possible
F	nodose (X)	CPM	11.90	9.86	0.13	4.63	9.1	Opr1	Oprm1	1.21x	1.15 to 1.28	1.000	4/4	pass	run
G	jugular (X)	CPM	9.62	8.33	0.92	0.90	7.2	Opr1	Oprm1	1.16x	0.92 to 1.46	0.898	2/3	pass	run
H	dorsal root	CPM	9.31	8.23	0.20	2.78	--	Opr1	Oprm1	1.13x	1.08 to 1.19	1.000	n/a	pass	not possible
I	celiac	CPM	25.03	0.42	0.03	0.56	66.4	Opr1	Oprk1	44.41x	20.42 to 197.63	1.000	1/1	pass	run
J	pelvic	CPM	16.33	0.50	0.43	0.46	51.0	Opr1	Oprm1	32.42x	22.00 to 54.20	1.000	2/2	pass	run
K	superior cervical	CPM	18.68	0.27	0.57	0.73	21.6	Opr1	Oprk1	25.72x	15.93 to 52.14	1.000	2/2	pass	run
L	lumbar chain	CPM	24.76	1.68	0.03	0.39	76.5	Opr1	Oprm1	14.78x	12.00 to 18.83	1.000	1/1	pass	run
M	thoracic chain	CPM	62.30	1.48	3.90	4.24	82.2	Opr1	Oprk1	14.69x	9.35 to 27.27	1.000	n/a	pass	not possible
N	stellate	CPM	27.55	0.56	0.03	4.53	52.5	Opr1	Oprk1	6.08x	5.16 to 7.27	1.000	1/1	pass	run
O	stellate	CPM	23.93	4.84	5.04	5.12	11.5	Opr1	Oprk1	4.68x	3.02 to 8.08	1.000	8/8	pass	not possible
P	sphenopalatine (VII)	CPM	19.50	0.29	0.04	0.01	49.9	Opr1	Oprm1	66.86x	40.54 to 132.54	1.000	1/1	pass	run
Q	intrinsic cardiac	CPM	11.13	3.04	0.36	0.00	38.2	Opr1	Oprm1	3.65x	3.29 to 4.08	1.000	3/3	pass	run
R	enteric submucosal, P24	CPM	35.57	0.18	16.14	3.77	32.4	Opr1	Oprd1	2.20x	2.05 to 2.37	1.000	3/3	pass	run
S	enteric submucosal, P7	CPM	26.70	0.74	2.29	122.51	23.4	Oprk1	Opr1	4.59x	4.04 to 5.24	1.000	2/2	pass	run

Why an ambient check could not run

- (A) deposit is 226 sorted neurons, no non-neuronal cells
- (B) deposit is 96 sorted neurons, no non-neuronal cells
- (D) deposit is Phox2b-sorted neurons, no non-neuronal cells
- (E) source h5ad is 0.6 GB and was not retained; not re-derived
- (G) scored on the pooled ganglion, not the jugular subset
- (H) source h5ad is 20.9 GB, above this session's disk allowance
- (M) deposit is 298 sorted neurons, no non-neuronal cells
- (O) deposit is author-filtered neurons, no non-neuronal cells